



Nonlinear dynamic modelling and analysis for a spur gear system with time-varying pressure angle and gear backlash

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ABSTRACT

In the previous nonlinear dynamic models of spur gear system, when considering bearing deformation, the pressure angle and gear backlash are usually considered as constants to facilitate the modelling. In fact, bearing deformation will cause the gear translation motion, and then lead to the changes of a series of gear mesh parameters. In view of this, this paper presents a new nonlinear dynamic model for a spur gear system considering the time-varying behaviors of pressure angle and gear backlash, together with the effects of gear gravity, unbalance mass, and internal/external excitations, etc. The equations of motion of the system are strictly derived using Lagrange method and solved by the numerical integral method. The time-varying mesh stiffness is calculated by the potential energy method based on an improved mesh stiffness model. Afterwards, the new model is compared with the previous model and some dynamic characteristic differences between the two models are demonstrated using the rotating speed and backlash as control parameters. This results shows that due to the effects of the time-varying pressure angle and backlash, the system may enter into chaotic motion in advance and the tooth impact state will also alter. This research can provide a useful source of reference for dynamic design of gear transmission system and the further study of spur gear system.

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1. Introduction

Gear mechanisms are widely used in many industrial applications, such as automobiles, wind turbines, ships, mining, aircrafts and other industries. With the increasingly strict restrictions on vibration and noise that may result from gear transmission systems, the prediction and control of gear's vibration and noise have attracted extensive attention. However, the mechanism of generating vibration and noise are very complex, and closely related to the dynamic characteristics of gear transmission systems. Therefore, it is essential to establish a relatively accurate mathematical model to gain some important dynamic characteristics for geared systems.

In the past decades, numerous mathematical models for geared system were proposed for the different kinds of purposes and many significant results have been obtained. Özgüven and Houser [1] comprehensively reviewed the linear vibration characteristics of geared systems. Wang [2] reviewed the mathematical models and the solving methods for the nonlinear dynamics of geared systems in detail and discussed the critical issues for further research on the nonlinear vibration in gear transmission systems. In earlier studies, Özgüven and Houser [3] used a single degree of freedom (SDOF) nonlinear

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model for the dynamic analysis of a gear pair including the effects of time-varying mesh stiffness, damping, transmission errors, profile modifications and backlash. Subsequently, Özgüven [4] developed a six-DOF nonlinear model for the dynamic analysis of spur gears including shaft and bearing dynamics and studied the effect of lateral-torsional coupled vibration on the dynamics of gears. Kahraman and Singh [5–7] proposed a series of studies in the influence of several factors on the frequency response characteristics of nonlinear geared system based on the SDOF and three-DOF models, such as time-variant mesh stiffness, backlash, bearing clearances, and external/internal excitations. Their results have laid a solid foundation for the following work. Blankenship et al. [8,9] examined analytically and experimentally nonlinear dynamic behavior of a mechanical oscillator with periodically time-varying parameters and clearance. Theodossiades and Natsiavas [10] applied a new analytical methodology to determine periodic steady state motions and their stability properties for a simple gear-pair system. Al-shyyab and kaharman [11,12] proposed a nonlinear time-varying dynamic model of a typical multi-mesh gear train and investigated the period-one, sub-harmonic and chaotic motions by using multi-term Harmonic Balance Method in conjunction with discrete Fourier Transforms. Shen et al. [13] used the incremental harmonic balance method to analyze the nonlinear dynamics of a spur gear pair including the backlash, time-varying stiffness and static transmission error. Moradi and Salarieh [14] investigated the nonlinear oscillation of a SDOF model of spur gear pairs in terms of dynamic transmission error (DTE) using multi-scale method, where the backlash nonlinearity is approximated with a third-order polynomial of DTE.

In recent years, with the development of nonlinear dynamics theories and numerical calculation, in addition to the effects of constant backlash and time-varying mesh stiffness, many other factors, such as bearing nonlinear force, shaft elastic deformation, gear eccentricity, and dynamic backlash, are also taken into account to make the mathematic model be closer to the realistic physical model. Kim et al. [15] proposed a new dynamic model for a pair of spur gears with time-varying pressure angle and contact ratio caused by gear translational motions due to bearing deformations. Subsequently, they [16] again extended their study to a planetary gear, but they don't consider the backlash nonlinearity in this models. Chen et al. [17] established the relationship between gear central distance and backlash and investigated the effects of the friction and dynamic backlash on the gear nonlinear dynamics, but the pressure angle is still a constant. Chen et al. [18] employed the fractal theory to express the backlash, and compared to other two ways of setting backlash as a fixed value or a steady random number that meets the normal distribution. Zhou et al. [19] investigated the dynamic behaviors of the spur-gear rotor bearing system considering the effects of the gear gravity, eccentricity, bearing clearance, transmission error, and coupled lateral-torsional vibration. Liu et al. [20] proposed a 10-DOF lateral-torsional-rocking coupled nonlinear dynamic model of a single gear system considering geometric eccentricity, gyroscopic moment, teeth to the partial load torque and dynamic backlash, and studied stiffness of the meshing model considering actual motion state and tooth profile modification using an analytical method. Xiang and Gao [21] investigated the effects of gear eccentricity and rotational speed on a gear-rotor-rolling bearing transmission system in detail. Cao et al. [22] proposed a new dynamic model of a planetary gear set with gear eccentricity errors and analyzed the effects of gear eccentricities on the dynamic performance of the system.

Although some nonlinear factors have been incorporated into dynamic analysis of gear system, less research has considered simultaneously the time-variant behaviors of both pressure angle and backlash in the nonlinear dynamic modelling. For convenience of modelling, most previous studies have treated the pressure angle and gear backlash as constants in nonlinear dynamic models. In fact, when a gear pair has the translational motions caused by bearing deformations, gear center distance will change and then result in the change of line of action (LOA), pressure angle, contact ratio and gear backlash. On the basis of Kim's research [15], this paper presents a new nonlinear dynamic model for a spur gear system considering the time-varying behaviors of pressure angle and backlash, together with the effects of the gear gravity, unbalance mass and internal/external excitations. The differential equations of motion of the system are derived strictly and this research can provide a useful reference for dynamic design of gear transmission system and the further study for spur gear system.

The remainder of this paper is organized as follows: Section 2 presents a six-DOF nonlinear dynamic model of a spur gear system, in which the time-varying mesh stiffness and backlash are given, respectively. In Section 3, the differential equations of motion of the gear system are derived based on Lagrange equation. In Section 4, the differences between the new and previous models are discussed using the rotating speed and backlash as control parameters. Finally, Section 5 gives some brief conclusions.

2. Dynamic model of a spur gear system

A six-DOF lumped parameter model of a pair of involute spur gear transmission system arranged horizontally is illustrated in Fig. 1. In this model, the shafts, pinion and gear, except for the teeth, are assumed to be rigid, and the shafts are supported by a pair of flexible ball bearings that can be modeled as linear springs and viscous dampers. Dynamic analysis is made in the plane of gear pair, and any out-of-plane motion is neglected. Here, m_i is the mass of gear i ($i = 1, 2$); J_i is the moment of inertia with respect to the mass center of gear i ; ω_i is the nominal angular velocity of gear i ; T_i is the torsional torque applied to gear i ; k_{xi} and k_{yi} ($i = 1, 2$) are the bearing stiffnesses in the x - y directions, respectively; c_{xi} and c_{yi} are the corresponding bearing dampings.

Fig. 2 describes the generalized coordinates for the mated gear pair. The motions of the gear set can be defined with six generalized coordinates as they are assumed to be planar motion, namely, two translational coordinates and one rotational coordinate for each gear. The dashed and solid circles represent the gear pair before and after motion, respectively. The axis

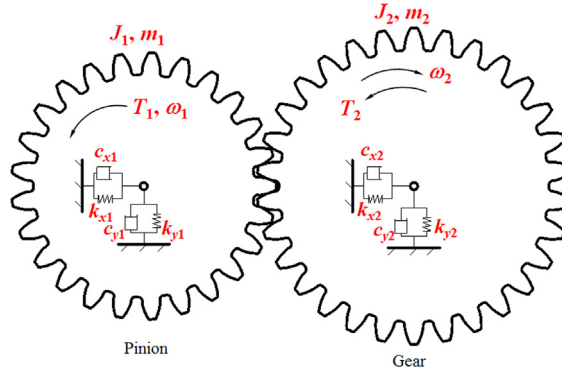


Fig. 1. Lumped-parameter model of spur gear pair.

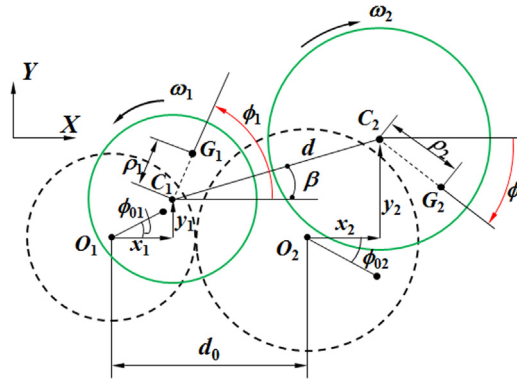


Fig. 2. Generalized coordinates for the gear pair.

centers of the static gears are denoted by points O_1 , O_2 , and the initial center distance is d_0 . Considering gear translational motions, the axis centers can move to points C_1 and C_2 , respectively. Here, two static coordinate systems $O_i x_i y_i$ ($i = 1, 2$) are established on the axis centers O_1 and O_2 , respectively. The angle β represents the position of the gear relative to the pinion after motion. G_i represents the mass center of gear i . The eccentricities of the two gears are denoted by ρ_1 and ρ_2 . ϕ_{01} and ϕ_{02} are the initial eccentricity angles. For simplicity, they are generally assumed to be zeroes at $t = 0$ s. The vibration displacements of gear i are given by translational coordinates, x_i and y_i . The rotation angle displacements of gear i are given by angular coordinates, ϕ_1 and ϕ_2 , which can be expressed as follows:

$$\phi_1(t) = \omega_1 t + \theta_1(t) + \phi_{01}, \phi_2(t) = \omega_2 t + \theta_2(t) + \phi_{02} \quad (1)$$

where, θ_i is the torsional angle displacement superimposed on the rigid-body rotation of gear i .

Let the rotation direction of pinion is positive, then the displacement vectors of the mass centers G_1 , G_2 in the coordinate systems $O_1x_1y_1$, $O_2x_2y_2$ can be expressed as, respectively.

$$\mathbf{r}_1 = (x_1 + \rho_1 \cos \phi_1) \mathbf{i} + (y_1 + \rho_1 \sin \phi_1) \mathbf{j}, \mathbf{r}_2 = (x_2 + \rho_2 \cos \phi_2) \mathbf{i} + (y_2 - \rho_2 \sin \phi_2) \mathbf{j} \quad (2)$$

where, $\mathbf{i}$ and $\mathbf{j}$ are unit vectors along the x and y axes, respectively.

Due to the translational motions of pinion and gear, the gear center distance d_0 is changed to d after the motion. The actual center distance d is given by

$$d = \sqrt{(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2} \quad (3)$$

The position angle β can be expressed as:

$$\beta = \tan^{-1} \frac{y_2 - y_1}{x_2 - x_1 + d_0} \quad (4)$$

Fig. 3 shows a nonlinear dynamic model for the gear pair at any mesh position after motion. The gear mesh is represented as a pair of rigid disks connected by a spring damper set along the LOA. The LOA is defined as the common tangent line of the base circles in the gear set having involute profiles. The actual pressure angle α is defined as the angle between the LOA and the direction of velocity at the pitch point, and can be expressed as:

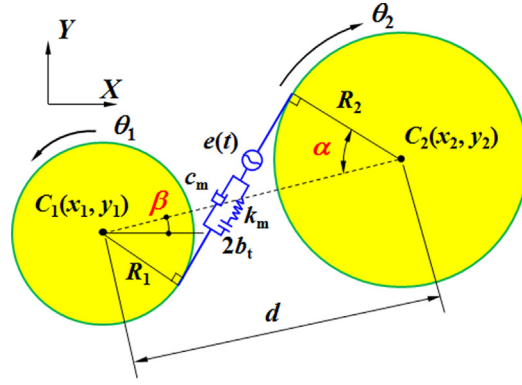


Fig. 3. Nonlinear dynamic model for the gear pair.

$$\alpha = \cos^{-1} \frac{R_1 + R_2}{d} = \cos^{-1} \frac{R_1 + R_2}{\sqrt{(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2}} \quad (5)$$

where, R_i is the radius of the base circle of gear i . It can be seen from Eqs. (4) and (5) that the gear position angle and pressure angle are affected by the translational motions of the gears while the previous studies nearly neglected these effects.

The change of gear center distance will also lead to the change of the gear contact ratio, according to the definition of the contact ratio, the actual contact ratio m_p can be given by

$$m_p = \frac{[\sqrt{R_{a1}^2 - R_1^2} + \sqrt{R_{a2}^2 - R_2^2} - d \sin \alpha]}{p_b} \quad (6)$$

here R_{a1} and R_{a2} are the radii of the addendum circles for pinion and gear, and p_b is the base pitch.

According to the gear mesh relationship, the gear mesh deformation (also called DTE) between two mating gears along the LOA can be written as [15,20,21]:

$$\delta(t) = (x_1 - x_2) \sin(\alpha - \beta) + (y_1 - y_2) \cos(\alpha - \beta) + R_1 \theta_1 - R_2 \theta_2 - e(t) \quad (7)$$

where, the static transmission error $e(t)$ is a displacement excitation related to the manufacturing and installation accuracy of gears. It can usually be approximated as a harmonic function: $e(t) = e_a \sin(\omega_m t)$, in which e_a is the error fluctuation and ω_m is the gear meshing frequency ($\omega_m = \omega_1 z_1$, z_1 is the number of teeth of pinion).

Based on the theory of viscoelasticity, the dynamic mesh force (DMF) consists of elastic force and damping force along the LOA and can be written as:

$$F_m = k_m f(\delta, b_t) + c_m f_1(\delta, b_t) \quad (8)$$

where, k_m and c_m are time-varying mesh stiffness and damping, $c_m = 2\xi_m \sqrt{k_m J_1 J_2 / (J_2 R_1^2 + J_1 R_2^2)}$, ξ_m is the damping ratio (0.03~0.17); b_t is time-varying half gear backlash. $f(\delta, b_t)$ and $f_1(\delta, b_t)$ are the nonlinear functions for backlash and relative speed respectively and can be expressed as:

$$f(\delta, b_t) = \begin{cases} \delta - b_t & (\delta > +b_t) \\ 0 & (|\delta| \leq b_t) \\ \delta + b_t & (\delta < -b_t) \end{cases} \quad (9)$$

$$f_1(\delta, b_t) = \begin{cases} \dot{\delta} - \dot{b}_t & (\delta > +b_t) \\ 0 & (|\delta| \leq b_t) \\ \dot{\delta} + \dot{b}_t & (\delta < -b_t) \end{cases} \quad (10)$$

here, superscript $(\cdot)$ denotes derivative with respect to time.

2.1. Calculation of time-varying mesh stiffness

In this paper, time-varying mesh stiffness k_m is calculated by the potential energy method based on an improved mesh stiffness model for a healthy gear pair, where gear tooth is simplified as a cantilever beam on the root circle, as shown in

Fig. 4. Besides, both the misalignment between the root circle and base circle and the accurate transition curve are also considered (Further details of the model can be found in Ref. [23].)

The time-varying mesh stiffness is a periodic excitation with the contact ratio m_p ($m_p < 2$) and a mesh period T_m ($T_m = 2\pi/\omega_m$). The contact ratio is closely related to the variation in gear mesh stiffness, because the proportions of the double and single tooth contact zones are determined by the contact ratio m_p , as shown in Fig. 5. For any integer n ($n \geq 1$), the double teeth contact zone is from $(n-1)T_m$ to $(m_p + n-2)T_m$, and k_d is the maximum mesh stiffness in this mesh process. The single tooth contact zone is from $(m_p + n-2)T_m$ to nT_m , and k_s is the minimum mesh stiffness. For simplicity, this study ultimately adopts time-varying mesh stiffness with a rectangular wave form. In other words, k_d and k_s represent the mesh stiffnesses in the double and single tooth contact zones, respectively.

2.2. Time-varying gear backlash

To meet the demand for good lubrication and simultaneously prevent jamming of the teeth due to tooth deformation and thermal expansion, a certain amount of gear backlash must be reserved between the non-working tooth flanks. In practice, backlash in gear transmission system can be divided into two parts: constant backlash and time-varying backlash. This part resulting from tooth thickness deviation and installment center distance error is constant backlash, which is also referred to as the initial backlash or design backlash denoted by $2b_0$ in this paper, and the other part caused by the gear center distance variation due to gear geometric eccentricity or bearing elastic deformation is time-varying backlash [24]. However, the time-variant behavior of backlash is usually neglected in the previous nonlinear dynamic models. Based on the meshing principle of involute gear, the dynamic backlash Δb induced by the change of gear center distance can be derived as [20]:

$$\Delta b = 2(R_1 + R_2)(\text{inv}(\alpha) - \text{inv}(\alpha_0)) \quad (11)$$

where $\text{inv}(x)$ is the involute function, $\text{inv}(x) = \tan(x) - x$; α_0 denotes the initial pressure angle for the initial center distance d_0 . When the gear pair is perfectly mounted, α_0 is generally equal to 20° .

Eventually, the total gear backlash $2b_t$ including the initial backlash $2b_0$ can be expressed as:

$$2b_t = 2b_0 + \Delta b = 2b_0 + 2(R_1 + R_2)(\text{inv}(\alpha) - \text{inv}(\alpha_0)) \quad (12)$$

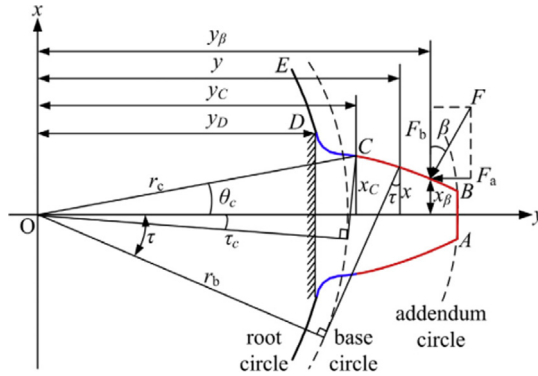


Fig. 4. Geometric model of the tooth profile.

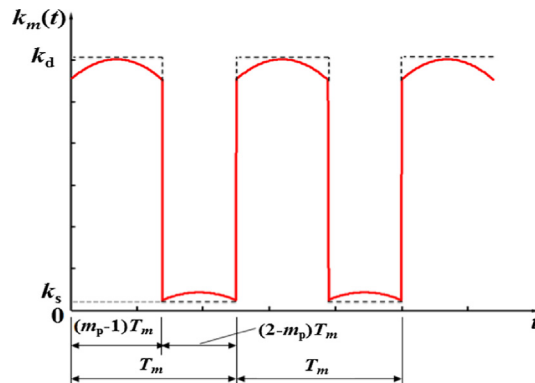


Fig. 5. Time-varying mesh stiffness.

It is obvious that the backlash is a function of the pressure angle α . Note that the backlash might be negative after motion, the real backlash is non-negative due to the existence of bearing clearance, so the initial backlash should be chosen reasonably.

3. Derivation of the equations of motion

The differential equations of motion for the gear pair can be derived by the six generalized coordinates, whose vector form is written as

$$\mathbf{q} = [x_1 \ y_1 \ \theta_1 \ x_2 \ y_2 \ \theta_2]^T \quad (13)$$

The kinetic energy T , potential energy U and dissipation function R of the system in terms of the generalized coordinates can be deduced, respectively.

Since the gear set has in-plane motion, the total kinetic energy T can be expressed as

$$T = m_1(\|\dot{\mathbf{r}}_1\|)^2 + J_1(\omega_1 + \dot{\theta}_1)^2 + m_2(\|\dot{\mathbf{r}}_2\|)^2 + J_2(\omega_2 + \dot{\theta}_2)^2/2. \quad (14)$$

The potential energy U due to bearing and tooth deformations can be written as

$$U = \frac{[(k_{x1}x_1^2 + k_{y1}y_1^2 + k_{x2}x_2^2 + k_{y2}y_2^2) + k_m f^2(\delta, b_t)]}{2}. \quad (15)$$

The total dissipation function R consists of the dissipation functions of the radial bearing damping and gear mesh damping.

$$R = [(c_{x1}\dot{x}_1^2 + c_{y1}\dot{y}_1^2 + c_{x2}\dot{x}_2^2 + c_{y2}\dot{y}_2^2) + c_m \dot{f}_1^2(\delta, b_t)]/2. \quad (16)$$

Considering the effect of gravity, the generalized force vector $\mathbf{Q}$ for the gear system are given by

$$\mathbf{Q} = [0 - m_1 g T_1 0 - m_2 g - T_2]^T \quad (17)$$

The differential equations of motion are derived using Lagrange's equation, which is given by

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} + \frac{\partial R}{\partial \dot{q}_j} = Q_j \quad (18)$$

Substitute Eqs. (14)–(17) into Eq. (18), the equations of motion of the system can be obtained as.

$$\begin{cases} m_1(\ddot{x}_1 - \rho_1 \ddot{\theta}_1 \sin \phi_1 - \rho_1 \dot{\phi}_1^2 \cos \phi_1) + c_{x1} \dot{x}_1 + k_{x1} x_1 + c_m f_1(\delta, b_t) \cdot f_{1,\dot{x}_1} + k_m f(\delta, b_t) \cdot f_{x_1} = 0 \\ m_1(\ddot{y}_1 + \rho_1 \ddot{\theta}_1 \cos \phi_1 - \rho_1 \dot{\phi}_1^2 \sin \phi_1) + c_{y1} \dot{y}_1 + k_{y1} y_1 + c_m f_1(\delta, b_t) \cdot f_{1,\dot{y}_1} + k_m f(\delta, b_t) \cdot f_{y_1} = -m_1 g \\ (J_1 + m_1 \rho_1^2) \ddot{\theta}_1 - m_1 \rho_1 (\ddot{x}_1 \sin \phi_1 - \ddot{y}_1 \cos \phi_1) + c_m f_1(\delta, b_t) \cdot f_{1,\dot{\theta}_1} + k_m f(\delta, b_t) \cdot f_{\theta_1} = T_1 \\ m_2(\ddot{x}_2 - \rho_2 \ddot{\theta}_2 \sin \phi_2 - \rho_2 \dot{\phi}_2^2 \cos \phi_2) + c_{x2} \dot{x}_2 + k_{x2} x_2 + c_m f_1(\delta, b_t) \cdot f_{1,\dot{x}_2} + k_m f(\delta, b_t) \cdot f_{x_2} = 0 \\ m_2(\ddot{y}_2 - \rho_2 \ddot{\theta}_2 \cos \phi_2 + \rho_2 \dot{\phi}_2^2 \sin \phi_2) + c_{y2} \dot{y}_2 + k_{y2} y_2 + c_m f_1(\delta, b_t) \cdot f_{1,\dot{y}_2} + k_m f(\delta, b_t) \cdot f_{y_2} = -m_2 g \\ (J_2 + m_2 \rho_2^2) \ddot{\theta}_2 - m_2 \rho_2 (\ddot{x}_2 \sin \phi_2 + \ddot{y}_2 \cos \phi_2) + c_m f_1(\delta, b_t) \cdot f_{1,\dot{\theta}_2} + k_m f(\delta, b_t) \cdot f_{\theta_2} = -T_2 \end{cases} \quad (19)$$

where, the comma denotes partial differentiation, e.g., $\delta_{x_1} = \frac{\partial \delta}{\partial x_1}$, $f_{x_1} = \frac{\partial f(\delta, b_t)}{\partial x_1}$, $f_{1,\dot{x}_1} = \frac{\partial f_1(\delta, b_t)}{\partial \dot{x}_1}$. It can be seen from Eq. (19) that the equations of motion are nonlinear and completely coupled to each other.

$$\dot{\delta}(t) = \dot{x}_1 \cdot \delta_{x_1} + \dot{y}_1 \cdot \delta_{y_1} + \dot{\theta}_1 \cdot \delta_{\theta_1} + \dot{x}_2 \cdot \delta_{x_2} + \dot{y}_2 \cdot \delta_{y_2} + \dot{\theta}_2 \cdot \delta_{\theta_2} - \dot{e}(t) \quad (20)$$

$$\dot{b}_t = (R_1 + R_2) \tan^2 \alpha \cdot (\dot{x}_1 \cdot \alpha_{x_1} + \dot{y}_1 \cdot \alpha_{y_1} + \dot{x}_2 \cdot \alpha_{x_2} + \dot{y}_2 \cdot \alpha_{y_2}) \quad (21)$$

$$\delta_{x_1} = \sin(\alpha - \beta) + [(x_1 - x_2) \cos(\alpha - \beta) - (y_1 - y_2) \sin(\alpha - \beta)](\alpha_{x_1} - \beta_{x_1}) \quad (22)$$

$$\delta_{y_1} = \cos(\alpha - \beta) + [(x_1 - x_2) \cos(\alpha - \beta) - (y_1 - y_2) \sin(\alpha - \beta)](\alpha_{y_1} - \beta_{y_1}) \quad (23)$$

$$\delta_{x_2} = -\sin(\alpha - \beta) + [(x_1 - x_2) \cos(\alpha - \beta) - (y_1 - y_2) \sin(\alpha - \beta)](\alpha_{x_2} - \beta_{x_2}) \quad (24)$$

$$\delta_{y_2} = -\cos(\alpha - \beta) + [(x_1 - x_2) \cos(\alpha - \beta) - (y_1 - y_2) \sin(\alpha - \beta)](\alpha_{y_2} - \beta_{y_2}) \quad (25)$$

$$\delta_{,\theta_1} = R_1, \delta_{,\theta_2} = -R_2 \quad (26)$$

$$f_{,x_i} = \begin{cases} \delta_{,x_i} - b_{t,x_i} & (\delta > +b_t) \\ 0 & (|\delta| \leq b_t) \\ \delta_{,x_i} + b_{t,x_i} & (\delta < -b_t) \end{cases}; \quad b_{t,x_i} = (R_1 + R_2) \tan^2 \alpha \cdot \alpha_{,x_i} (i = 1, 2) \quad (27)$$

$$f_{,y_i} = \begin{cases} \delta_{,y_i} - b_{t,y_i} & (\delta > +b_t) \\ 0 & (|\delta| \leq b_t) \\ \delta_{,y_i} + b_{t,y_i} & (\delta < -b_t) \end{cases}; \quad b_{t,y_i} = (R_1 + R_2) \tan^2 \alpha \cdot \alpha_{,y_i} (i = 1, 2) \quad (28)$$

$$f_{,\theta_1} = \begin{cases} R_1 - b_{t,\theta_1} & (\delta > +b_t) \\ 0 & (|\delta| \leq b_t) \\ R_1 + b_{t,\theta_1} & (\delta < -b_t) \end{cases}, b_{t,\theta_1} = 0; f_{,\theta_2} = \begin{cases} -R_2 - b_{t,\theta_2} & (\delta > +b_t) \\ 0 & (|\delta| \leq b_t) \\ -R_2 + b_{t,\theta_2} & (\delta < -b_t) \end{cases}, b_{t,\theta_2} = 0 \quad (29)$$

$$f_{1,x_i} = f_{,x_i}; f_{1,y_i} = f_{,y_i}; f_{1,\theta_i} = f_{,\theta_i}; (i = 1, 2) \quad (30)$$

in which

$$\alpha_{,x_1} = -\alpha_{,x_2} = -\frac{(R_1 + R_2)(x_2 - x_1 + d_0)}{\left[(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2\right] \sqrt{(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2 - (R_1 + R_2)^2}} \quad (31)$$

$$\alpha_{,y_1} = -\alpha_{,y_2} = -\frac{(R_1 + R_2)(y_2 - y_1)}{\left[(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2\right] \sqrt{(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2 - (R_1 + R_2)^2}} \quad (32)$$

$$\beta_{,x_1} = -\beta_{,x_2} = \frac{y_2 - y_1}{(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2}, \quad \beta_{,y_1} = -\beta_{,y_2} = -\frac{x_2 - x_1 + d_0}{(x_2 - x_1 + d_0)^2 + (y_2 - y_1)^2} \quad (33)$$

However, in previous studies the pressure angle was often assumed not to be influenced by the translational motion, namely, the pressure angle and the contact ratio were regarded as constants and the position angle ψ was not introduced. Besides, gear backlash is set as a constant as well. In this case, the pressure angle α and the position angle β can be given by

$$\alpha = \cos^{-1} (R_1 + R_2)/d_0, \beta = 0 \quad (34)$$

And the gear mesh deformation Eq. (7) and its derivative Eq. (20) are re-defined as:

$$\delta(t) = (x_1 - x_2) \sin \alpha + (y_1 - y_2) \cos \alpha + R_1 \theta_1 - R_2 \theta_2 - e(t) \quad (35)$$

$$\dot{\delta}(t) = (\dot{x}_1 - \dot{x}_2) \sin \alpha + (\dot{y}_1 - \dot{y}_2) \cos \alpha + R_1 \dot{\theta}_1 - R_2 \dot{\theta}_2 - \dot{e}(t) \quad (36)$$

Thus, the equation of motion Eq. (19) can be simplified into the following equations set.

$$\begin{cases} m_1(\ddot{x}_1 - \rho_1 \ddot{\theta}_1 \sin \phi_1 - \rho_1 \dot{\phi}_1^2 \cos \phi_1) + c_{x1} \dot{x}_1 + k_{x1} x_1 + F_m \sin \alpha = 0 \\ m_1(\ddot{y}_1 + \rho_1 \ddot{\theta}_1 \cos \phi_1 - \rho_1 \dot{\phi}_1^2 \sin \phi_1) + c_{y1} \dot{y}_1 + k_{y1} y_1 + F_m \cos \alpha = -m_1 g \\ (J_1 + m_1 \rho_1^2) \ddot{\theta}_1 - m_1 \rho_1 (\ddot{x}_1 \sin \phi_1 - \ddot{y}_1 \cos \phi_1) + F_m = T_1 \\ m_2(\ddot{x}_2 - \rho_2 \ddot{\theta}_2 \sin \phi_2 - \rho_2 \dot{\phi}_2^2 \cos \phi_2) + c_{x2} \dot{x}_2 + k_{x2} x_2 - F_m \sin \alpha = 0 \\ m_2(\ddot{y}_2 - \rho_2 \ddot{\theta}_2 \cos \phi_2 + \rho_2 \dot{\phi}_2^2 \sin \phi_2) + c_{y2} \dot{y}_2 + k_{y2} y_2 - F_m \cos \alpha = -m_2 g \\ (J_2 + m_2 \rho_2^2) \ddot{\theta}_2 - m_2 \rho_2 (\ddot{x}_2 \sin \phi_2 + \ddot{y}_2 \cos \phi_2) - F_m = -T_2 \end{cases} \quad (37)$$

The dynamic mesh force can be expressed as:

$$F_m = k_m f(\delta, b_0) + c_m \dot{f}_1(\delta, b_0) \quad (38)$$

The backlash and speed nonlinear functions are given as follows:

$$f(\delta, b_0) = \begin{cases} \delta - b_0 & (\delta > +b_0) \\ 0 & (|\delta| \leq b_0) \\ \delta + b_0 & (\delta < -b_0) \end{cases} \quad (39)$$

$$f_1(\delta, b_0) = \begin{cases} \dot{\delta} & (|\delta| > b_0) \\ 0 & (|\delta| \leq b_0) \end{cases} \quad (40)$$

where b_0 is half the initial backlash.

When the gear's mass eccentricity isn't considered, the effect of gear gravity may be neglected in the gear dynamic models. Thus, the equation of motion Eq. (37) will be further simplified into a typical dynamic model shown in Fig. 6, which has often been applied in the previous studies.

$$\begin{cases} m_1 \ddot{x}_1 + c_{x1} \dot{x}_1 + k_{x1} x_1 + F_m \sin \alpha = 0 \\ m_1 \ddot{y}_1 + c_{y1} \dot{y}_1 + k_{y1} y_1 + F_m \cos \alpha = 0 \\ J_1 \ddot{\theta}_1 + F_m R_1 = T_1 \\ m_2 \ddot{x}_2 + c_{x2} \dot{x}_2 + k_{x2} x_2 - F_m \sin \alpha = 0 \\ m_2 \ddot{y}_2 + c_{y2} \dot{y}_2 + k_{y2} y_2 - F_m \cos \alpha = 0 \\ J_2 \ddot{\theta}_2 - F_m R_2 = -T_2 \end{cases} \quad (41)$$

4. Comparison and discussion

Eqs. (19) and (41) represent for the new model (with time-varying pressure angle and backlash) and the previous model (with constant pressure angle and backlash), respectively. From the above calculation and analysis, the new model is a very complex system with strong nonlinearity, time variance and coupling and whether it can predict the dynamic performances of the system better need to be analyzed. In this section, a case on high-speed level for a wind turbine gearbox is adopted to verify the differences between the new and previous models. In view of the stochastic nature of wind speed, the input and output torques exist some fluctuation and can be represented by first-order harmonic functions.

$$T_1 = T_{10} + T_{1a} \sin(\omega_1 t + \varphi_{10}), T_2 = T_{20} + T_{2a} \sin(\omega_2 t + \varphi_{20}) \quad (42)$$

where, T_{10} and T_{20} denote the mean torques; T_{1a} and T_{2a} indicate the fluctuations; φ_{10} and φ_{20} represent the initial phase angles.

The system models are solved by MATLAB's ODE15s function, which is very suitable for solving the stiff problem. To compare the dynamic characteristics between the new and previous models under the same parameters, in the following analysis, the influences of gear mass eccentricity and gravity are neglected, namely, the mass eccentricities ρ_1, ρ_2 and the gravity acceleration g are all assumed to be zeroes in Eq. (19). Then, some dynamic characteristics between the new and previous models are compared using the rotating speed and initial backlash as control parameters. The main parameters of the spur gear system are listed in Table 1.

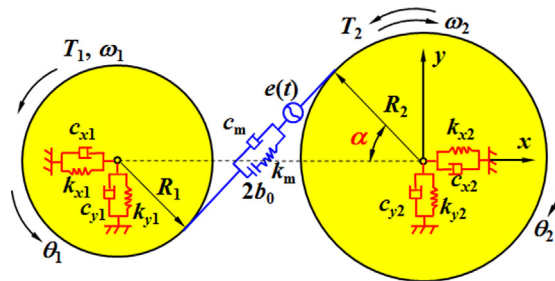


Fig. 6. Typical six-DOF nonlinear dynamic model of spur gear pair.

Table 1
Main parameters of the spur gear system.

Parameter/property	Value	Parameter/property	Value
Number of teeth z_1/z_2	20	Bearing stiffness $k_{x1, y1}, k_{x2, y2}$ (N/m)	1.0e8
Module m_n (mm)	10	Bearing damping $c_{x1, y1}, c_{x2, y2}$ (N·s/m)	512.64
Tooth width B (mm)	30	Error amplitude e_a (μ m)	20
Initial pressure angle α_0 ($^\circ$)	20	Torque mean T_{10}, T_{20} (N·m)	300
Initial center distance d_0 (mm)	200	Torque fluctuation T_{1a}, T_{2a} (N·m)	100
Designed contact ratio m_p	1.5568	Mesh damping ratio ξ_m	0.07
Gear mass m_1, m_2 (kg)	6.57	Mesh stiffness in single tooth contact k_s (N/m)	3.6228e8
Moment of inertia J_1, J_2 (kg·m ²)	0.0365	Mesh stiffness in double teeth contact k_d (N/m)	6.5072e8

4.1. Model validation

Although many dynamic models for spur gear system were presented for the different purposes, few studies have considered the time-varying pressure angle and backlash simultaneously in nonlinear dynamic model. For simplicity, the time variations of pressure angle and backlash in the previous model are usually neglected, which can lead to some different dynamic characteristics. In fact, when considering translational motions due to bearing deformation, the dynamic characteristics of geared system can be changed, especially at high speeds. Fig. 7 shows the time-domain curves of gear center distance, pressure angle, gear backlash and contact ratio when the rotating speed $n_1 = 1000$ r/min, the initial backlash

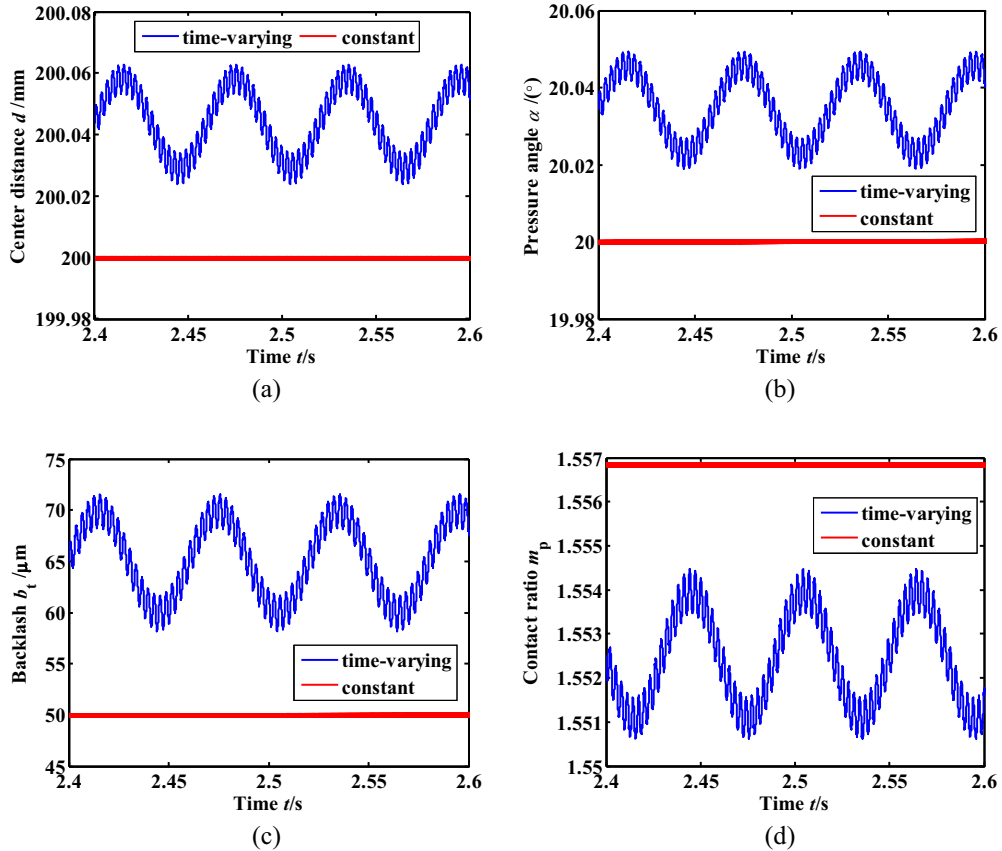


Fig. 7. Time-domain curves of center distance, pressure angle, backlash and contact ratio at $n_1 = 1000$ r/min.

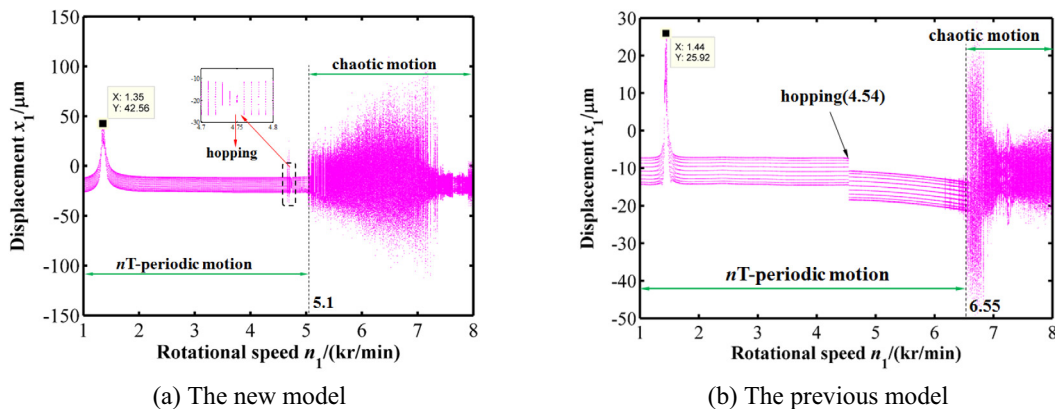


Fig. 8. Bifurcation diagrams of x_1 versus n_1 for the new and previous models.

$b_0 = 50 \mu\text{m}$ and the other parameters keep unchanged. Because of the coupling effects between gears' meshing parameters and vibration responses, the gear center distance, pressure angle, gear backlash and contact ratio will change with time in gear transmission. As can be seen from Fig. 7, the gear center distance fluctuates around 200.044 mm and is larger than the theoretical value 200 mm ; the pressure angle changes around 20.03° and is greater than the ideal value 20° ; the backlash waves around $64.98 \mu\text{m}$ and is much more than the initial value $50 \mu\text{m}$; however, the contact ratio varies around 1.5525 and is less than the design value 1.5568 . Through the analysis, the gear center distance variation has a significant influence on the backlash, which will affect the nonlinear characteristics of the system. Thus, the proposed new model in this study is more realistic than the previous model to a certain extend.

4.2. Effects of rotating speed of pinion on dynamic responses

Rotating speed of pinion is one of the key parameters that affect the dynamic behaviors of the mechanical transmission system and commonly used as control parameter to investigate the nonlinear dynamic characteristics of the system.

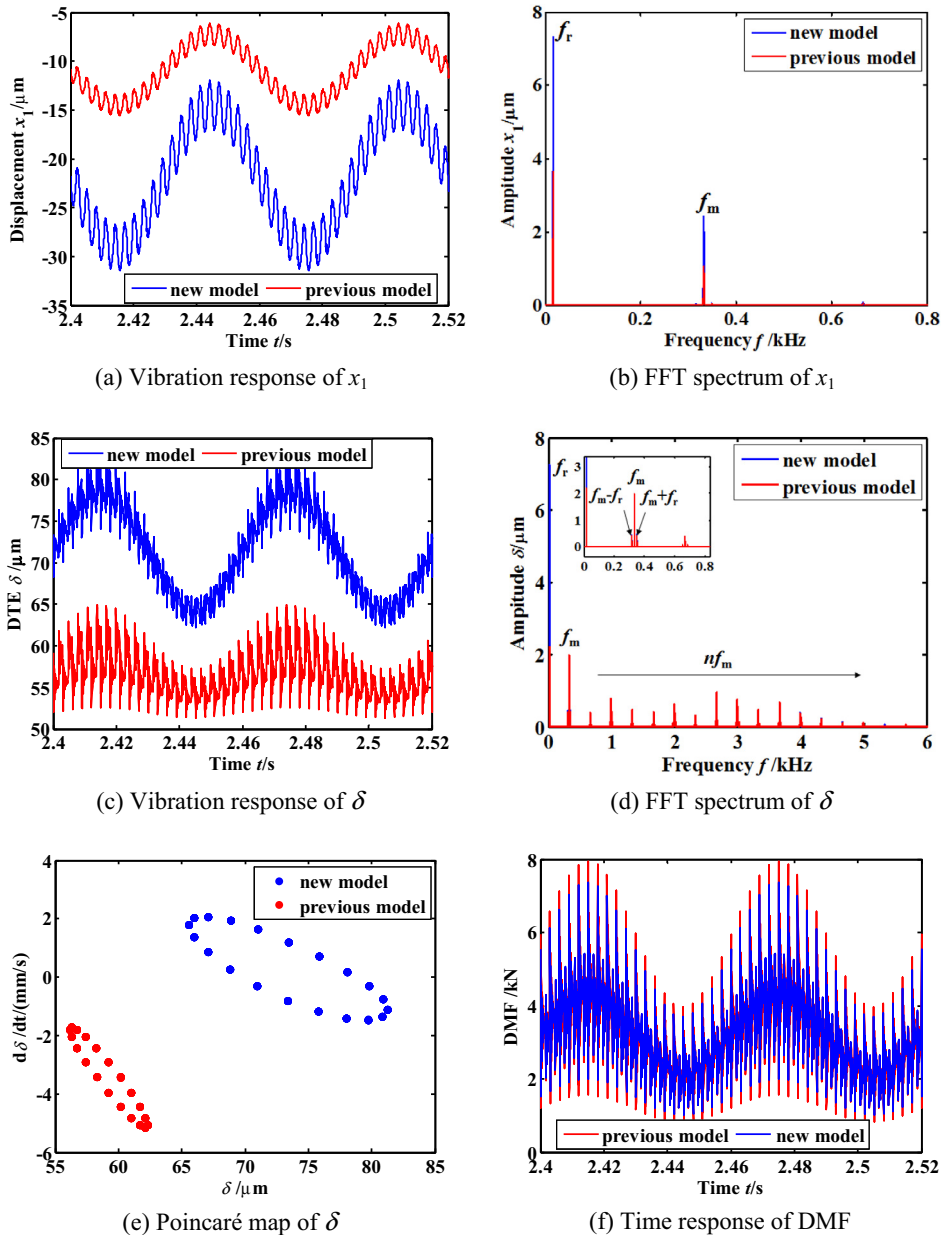


Fig. 9. Comparison of dynamic responses at low speed $n_1 = 1000 \text{ r/min}$.

Accordingly, in this section, we will first compare the global characteristics by using bifurcation diagrams with respect to the rotating speed n_1 between the new and previous models, and then further discuss the differences of the predicted responses with the two models under the different speeds. Here, let the initial backlash b_0 is equal to 50 μm and the other parameters are the same as Table 1. The bifurcation diagrams of the maximum of vibration response x_1 with the change of n_1 are shown in Fig. 8.

Fig. 8 shows that both nonlinear models have some common ground on the prediction of the bifurcation behavior and resonance of the system. The two cases both induce the crisis route to chaotic motion. But, the new model system enters into chaotic motion in advance. By comparing the bifurcation diagrams, it is found that the new model produces the resonance at $n_1 = 1.35$ kr/min prior to the previous model at $n_1 = 1.44$ kr/min. Besides, there exist ‘frequency hopping’ phenomena in each bifurcation diagram, which will cause the fluctuation of dynamic load between the teeth.

To understand the differences between the new and previous models more deeply, the dynamic responses of the system at $n_1 = 1000$ r/min, 3000 r/min, 4500 r/min and 6000 r/min are taken for examples, as shown in Figs. 9–12. The dynamic

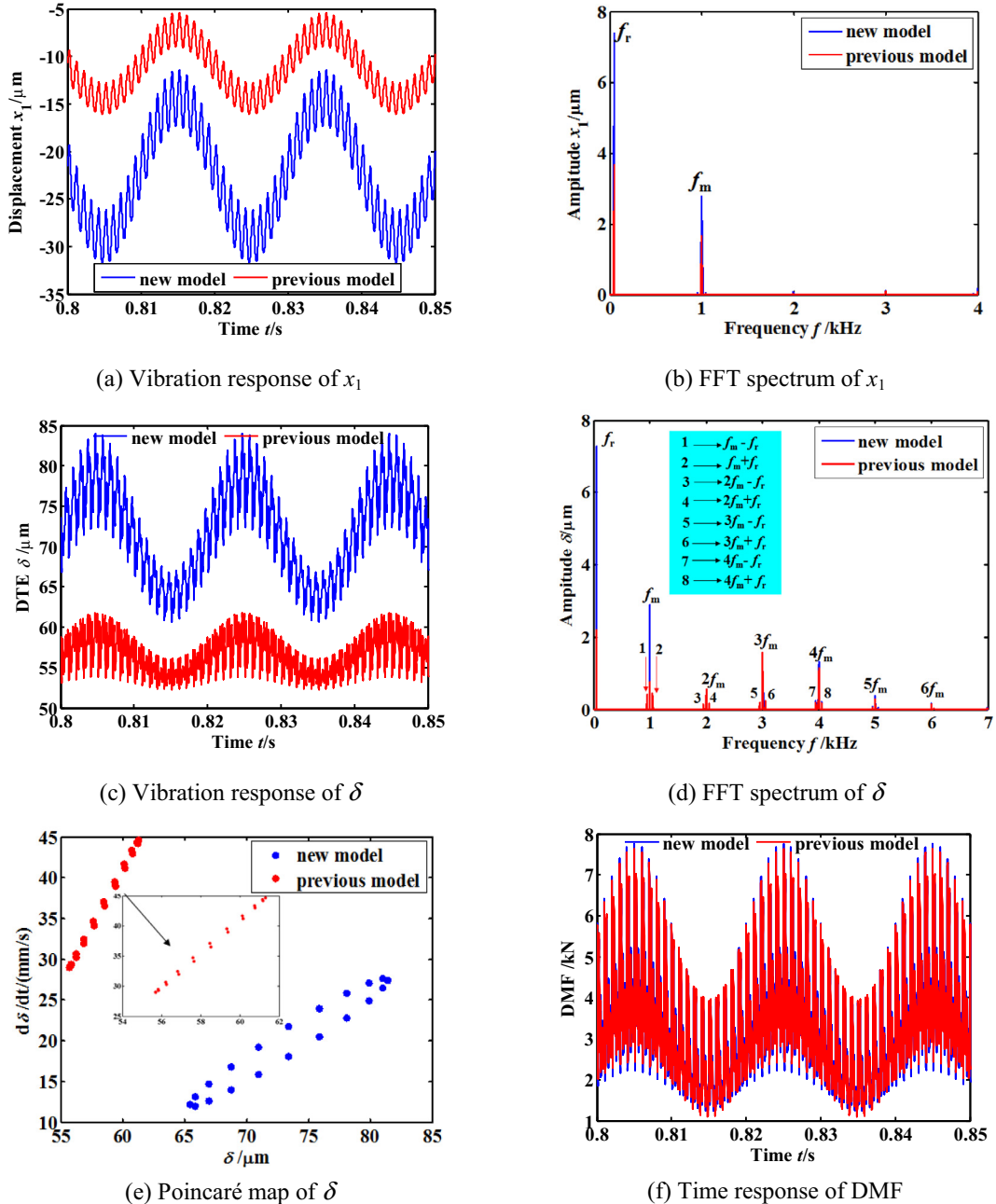


Fig. 10. Comparison of dynamic responses at medium speed $n_1 = 3000$ r/min.

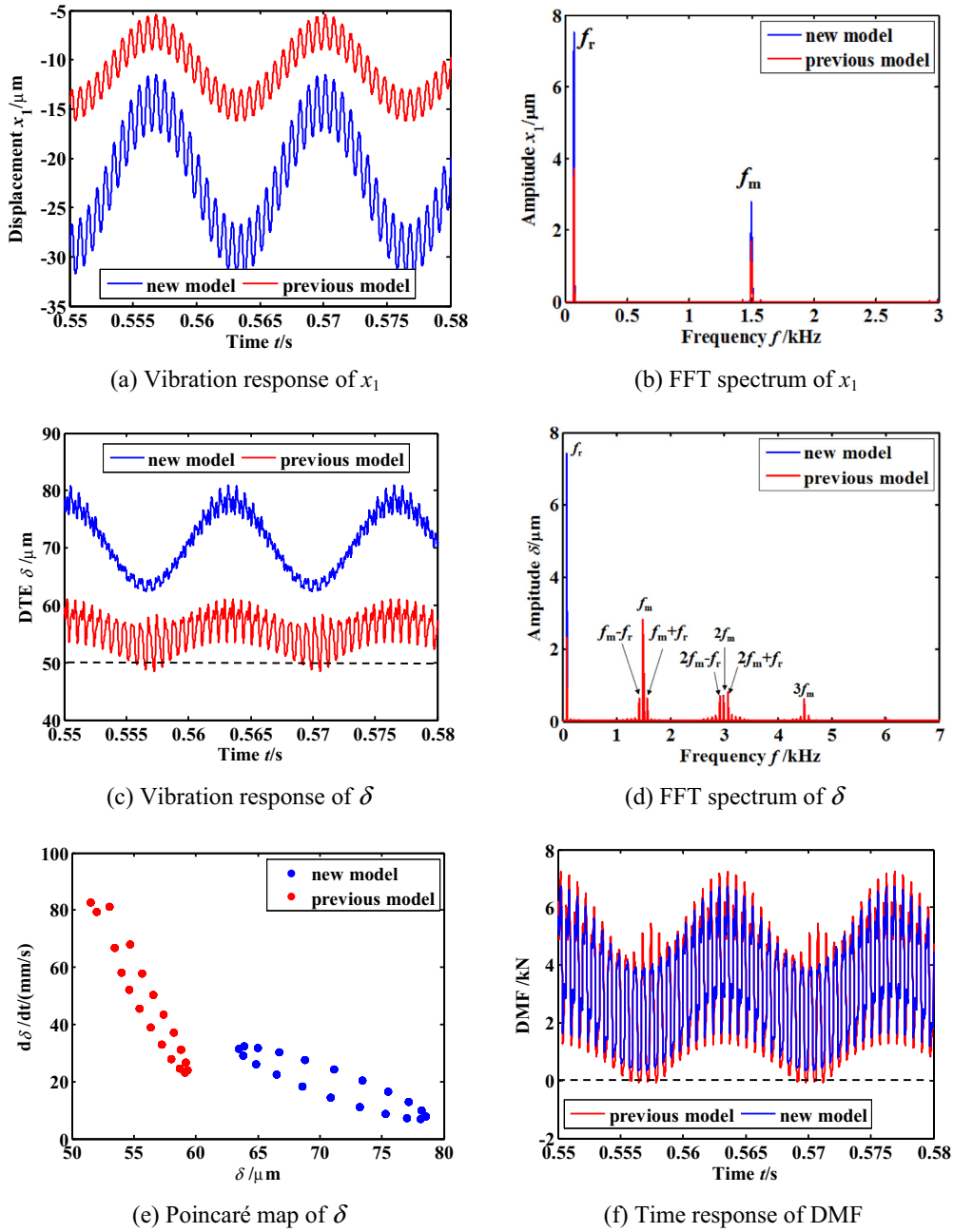


Fig. 11. Comparison of dynamic responses at high speed $n_1 = 4500\text{r/min}$.

responses of the system at low speed $n_1 = 1000\text{ r/min}$ is shown in Fig. 9. As can be seen from Fig. 9(a and b), the amplitude of displacement x_1 of the new model is far larger than that of the previous model and the amplitude difference of x_1 between the two models mainly locates in the rotational frequency f_r ($f_r = n_1/60 = 16.67\text{ Hz}$) and the mesh frequency f_m ($f_m = z_1 f_r = 333.33\text{ Hz}$). From Fig. 9(c), the fluctuation of the DTE obviously intensifies when considering time-variant pressure angle and backlash. The f_r , frequency multiplication ($n \cdot f_m$) and combination frequency ($f_m \pm f_r$) components appear in FFT spectrum in Fig. 9(d) and the amplitude difference of the DTE between the two models mainly lies in the f_r . As is known, for the constant backlash model, it is easy to judge the teeth impact state from the relationship between the DTE and constant backlash. Due to $\delta_{\min} > 50\text{ }\mu\text{m}$, it can be concluded that the previous model system is in non-impact state. However, since the backlash of the new model is time-variant, it is difficult to determine the impact state of the system using the above means. At this time, the DMF between the teeth is a good choose to estimate the teeth impact state. From Fig. 9(f), the maximum of DMF of the new model is instead less than that of the previous model, which is not as we expect, namely, the

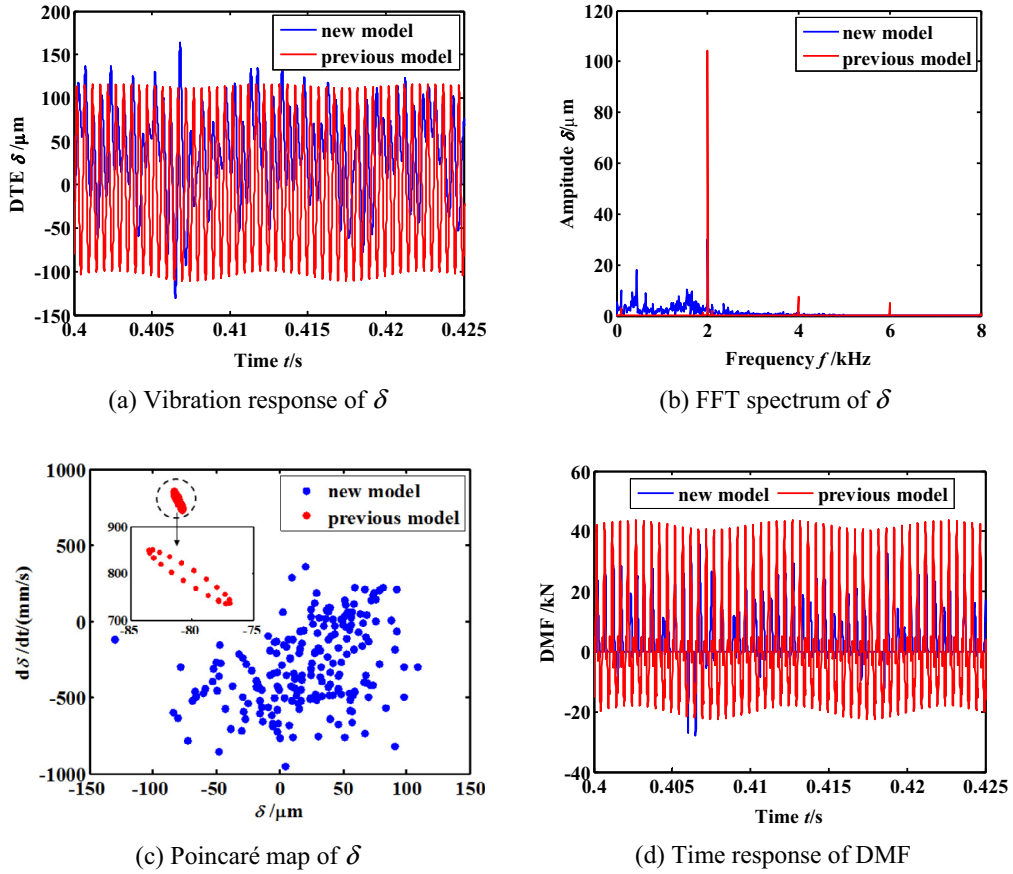


Fig. 12. Comparison of dynamic responses at high speed $n_1 = 6000$ r/min.

greater the DTE is, the greater the DMF should be. This is because the DTE and backlash of the new model increase simultaneously, as shown in Figs. 7(c) and 9(c). Furthermore, due to the minimum dynamic mesh force $F_{\min} > 0$ under the two models, the non-impact state of the system can also be verified further. In Fig. 9(e), there are 20 discrete return points in the Poincaré map for the two models, which reveals that the system responses are both period-20 motion.

When the system is operating at medium speed $n_1 = 3000$ r/min, the dynamic responses of the system under the two models are shown in Fig. 10. From Fig. 10(a,c), the amplitudes of displacement x_1 and DTE of the new model is still more than those of the previous model. There occur a variety of frequency components in the spectrum of DTE in Fig. 10(d) and the amplitude difference of DTE between the two models mainly locates in the f_r and f_m . The system still undergoes a period-20 motion under the two models in Fig. 10(e), but the maximum of DMF of the new model is slightly higher than that of the previous model, as indicated in Fig. 10(f). According to the minimum dynamic mesh force $F_{\min} > 0$, it can be inferred that the system under the two models lies in the non-impact state.

Figs. 11 and 12 show the dynamic response of the system at high speeds $n_1 = 4500$ r/min and 6000 r/min, respectively. As can be seen from Fig. 11(a,c), the amplitudes of displacement x_1 and DTE of the new model is still higher than those of the previous model. The maximum of DMF of the new model illustrated in Fig. 11(f) is less than that of the previous model, which is somewhat similar to Fig. 9(f). According to $F_{\min} = 0$ or $\delta_{\min} < 50 \mu\text{m}$ for the previous model, it can be inferred that the previous model undergoes the single-sided impact, but the new model has non-impact in terms of $F_{\min} > 0$, as shown in Fig. 11(c,f). Although the impact states between the two models are different, the system is still period-20 motion.

It can be seen from Fig. 12 that when considering time-varying pressure angle and backlash, the new model has entered into chaotic motion but the previous model still experiences period-20 motion. In addition, due to the dynamic mesh force $F_{\min} < 0$ and $F_{\max} > 0$, the system for the two models undergoes the double-sided impact, as shown in Fig. 12(d).

To further describe the change of the maximum dynamic mesh force $F_{\max}$ with the rotating speed, the dynamic load factor (DLF) is defined as follows:

$$DLF = \frac{F_{\max}}{F_s} = \frac{\max[F_m(t)]}{F_s} \quad (43)$$

where, $F_{\max}$ is considered as the maximum dynamic mesh force during simulation time, F_s is the static mesh force defined as the ratio of the average input torque to the base radius of the pinion, and F_m is the dynamic mesh force.

Fig. 12 shows the changing curves of DLFs with the pinion speed under the two models. It can be observed that the mutation positions of DLFs in Fig. 13 are corresponding to the resonance, hopping and bifurcation points in the bifurcation diagrams in Fig. 8, respectively. Additionally, the DLFs between the two models alternates in the periodic region $n_1 \in [1, 4.54]$ kr/min, which also shows that when the DTE of the new model is greater than that of the previous model, the maximum of DMF of the new model is not always greater than that of the previous model. This is because when the DTE for the new model increases, the backlash will increase simultaneously.

4.3. Effects of gear backlash on dynamic responses

As is known, gear backlash plays an important role in gear transmission system, but its existence can lead to the mesh impact and teeth separation phenomenon. The initial backlash b_0 can be adjusted by tooth thickness deviation and gear central distance. As mentioned before, gear backlash is actually time-varying instead of constant, and its increase or decrease can result in the different dynamic characteristics. Therefore, it is necessary to study the influence of the backlash on the dynamic characteristics of gear transmission system. Fig. 14 is the bifurcation diagrams of the maximum of vibration response x_1 with the initial backlash b_0 changes when n_1 value is equal to 6000r/min, and the other parameters remain the same as the previous section (Section 4.2.)

It can be seen from Fig. 14 that the system both first undergoes periodic motion and then enters into chaotic motion by the crisis route for the new and previous models. But, the new model enters into chaotic region before the previous model. Besides, the displacements x_1 of the two models linearly change with the increase of the b_0 in the periodic region.

Fig. 15 depicts the dynamic responses of the system for the new and previous models when $b_0 = 20 \mu\text{m}$. It is obvious that the oscillation amplitudes of displacement x_1 , DTE and DMF of the new model are all higher than those of the previous

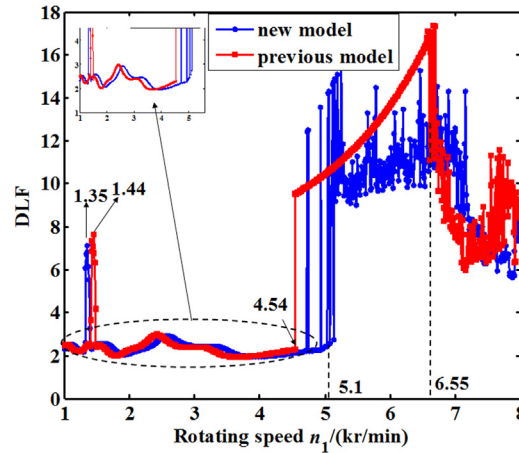


Fig. 13. Comparison of DLFs at the different rotating speeds.

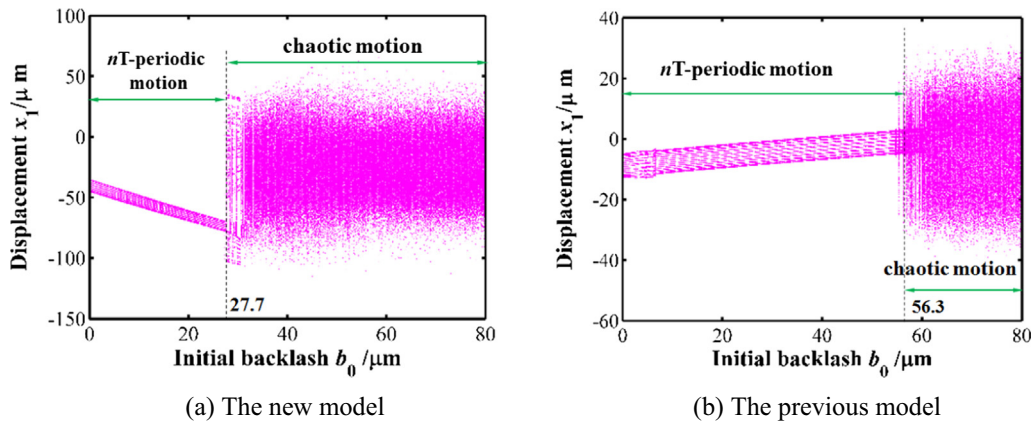


Fig. 14. Bifurcation diagrams of x_1 versus b_0 for the new and previous models when $n_1 = 6000\text{r/min}$.

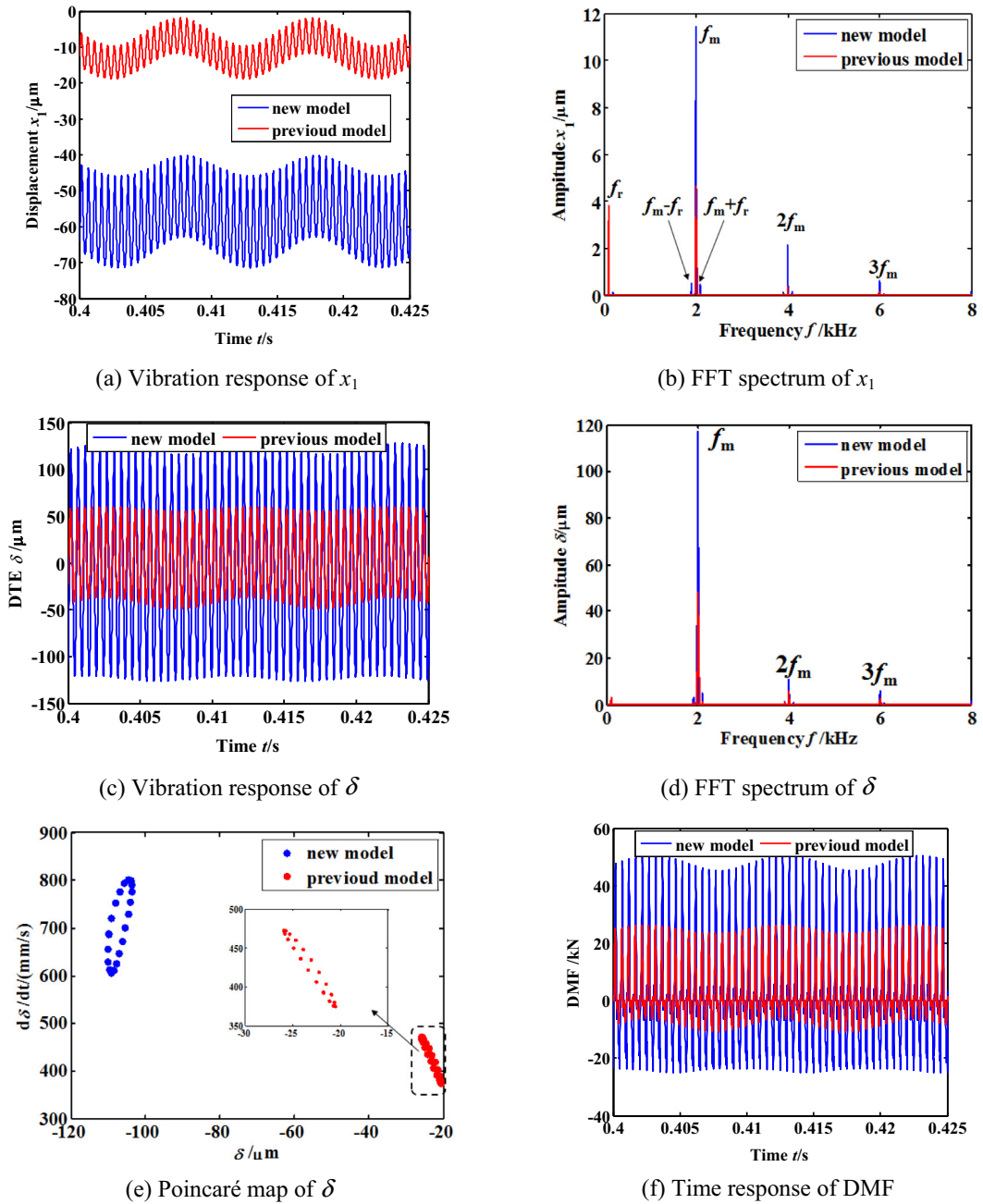


Fig. 15. Comparison of dynamic responses at $b_0 = 20 \mu\text{m}$.

model. The vibration responses both change periodically and the Poincaré map shows 20 discrete return points in Fig. 15(e). The results reveal that the system response is period-20 motion under the two models. As illustrated in Fig. 15(f), the system has the double-sided impact due to $F_{\max} > 0$ and $F_{\min} < 0$.

Fig. 16 shows the changes of the DLFs (indicating the maximum dynamic mesh force) with the initial backlash at high speed $n_1 = 6000 \text{ r/min}$. It can be seen that the DLFs of two models linearly increase during the periodic motion time. After the system entering into chaos, the changes of the DLFs appear very irregular, which are corresponding to the bifurcation diagrams in Fig. 14. Besides, in the common periodic region $b_0 \in [0, 27.7]$, the DLFs of the new model is much larger than those of the previous model. This implies the backlash variation will have a great effect on the system dynamic characteristics.

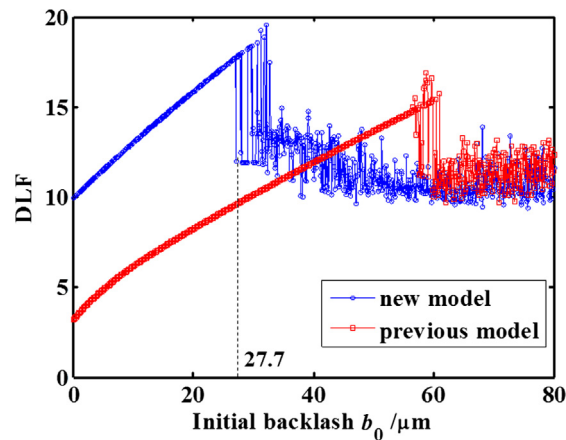


Fig. 16. Comparison of DLFs at the different backlashes.

5. Conclusions

This paper presents a new multi-DOF dynamic model for a spur gear system considering the time-varying behaviors of both pressure angle and backlash, together with the effects of the gear gravity, unbalance mass, and the internal and external excitations. The dynamic equations of the system are derived using Lagrange method and solved by the numerical integral method. With the help of bifurcation diagrams, displacement x_1 , DTE, DMF and DLF, the differences between the new and previous models are compared under the effects of two parameters: the rotating speed of pinion and the initial backlash. Some beneficial conclusions are obtained as follows:

- (1) The proposed new nonlinear dynamic model has time-varying pressure angle and gear backlash. Therefore, the new model can yield a more realistic dynamic response than the previous model does.
- (2) Both nonlinear models have some common ground on the prediction of the bifurcation behavior and resonance of the system. But, the system may enter into chaos motion in advance due to the effects of time-varying pressure angle and backlash.
- (3) Under a certain backlash, with the increase of the rotating speed, the amplitudes of displacement x_1 and DTE of the new model are much more than those of the previous model in the common periodic region. However, the maximum of DMF of the new model is not always greater than that of the previous model due to the dynamic backlash.
- (4) The initial backlash has a significant influence on the dynamic characteristics of the gear system. Under high speeds, with the increasing backlash, the dynamic load factor of the system linearly increases during the periodic motion time. Moreover, the DLFs of the new model are far higher than those of the previous model in the common periodic region, so are the maximum DMFs.

Acknowledgements

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